

An immersive interface for remote collaboration with multiple telepresence robots through digital twin spaces

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Abstract

A human-robot collaboration system “Telecobot” is proposed, in which a remote user allocates tasks to multiple telepresence robots via digital twin spaces. The digital twin spaces of local sites where robots work is used as a medium to observe the real spaces from a remote site and to give instructions to the smart robots on the local site. Telecobot supports a remote user understanding the real-time status of local sites distributed geographically and allocating tasks to each of the robots. We assume that the robots are smart enough to execute the series of tasks assigned by the remote user. This division of roles is a major characteristic of Telecobot, where a remote user is responsible for judging the situation and building a series of tasks via the digital twin space, while multiple robots working in each of the local sites executes the individual tasks following the information built by the remote user in the digital twins of the local sites. An experiment assuming telecollaboration with smart robots working at nursing care sites demonstrated the division of roles for the care works and showed the potential of increasing the productivity of the work by using the Telecobot interface.

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1 Introduction

The nursing care industry is facing labor shortage, with an estimated supply-demand gap of approximately 377,000 workers as of 2025 in Japan [Persol Research Institute, 2018]. The use of robots is an effective way to fill this gap. Sugano defines five levels to be achieved by smart robots and projected the year for achieving

the highest level 5, fully independent (robots perform tasks completely independently without human instructions), will be after 2050 [Kanno, 2021]. The development of autonomous robots that are not fully independent but are designed to cooperate with humans is a promising approach to deal with the labor shortage for the decades ahead in the future.

There are two types of human robot interaction. The first is the division of labor between humans and robots at the same site, which is called human-robot collaboration. The second is remote robot operation addressed in this study [Taichi et al., 2020][Yigitbas et al., 2020]. Therefore, instead of one person running one robot, a system should be built in which one person distributes tasks to multiple robots.

2 IMMERSIVE INTERFACE OF TELECOBOT

Figure 1 shows the case of instructing two robots using the immersive interface of Telecobot by an operator in a remote site through the two digital twin spaces of the local sites where each of the robots work. The steps for the instruction for the system are as follows.

- A local site is reconstructed in three dimensions in a digital twin space composed of point cloud data.
- The point cloud is classified into objects in the local site to be analyzed for building tasks.
- The remote operator is immersed in the digital twin space and observes the local site and the robot model to understand the real time situation. (Figure2)
- The operator gives instructions to each of the robots on site by editing the point cloud or adding virtual objects in the digital twin space to visualize the state after the task execution.
- The robot receives the tasks indicated as differences in the digital twin space.
- The robot understands the tasks and executes them in sequence and the updated situation of the local site is reflected in the digital twin space.

When a robot is executing tasks, the remote operator can give tasks to another robot in another local site. This system interface of Telecobot allows the operator to operate multiple remote robots working in multiple sites.

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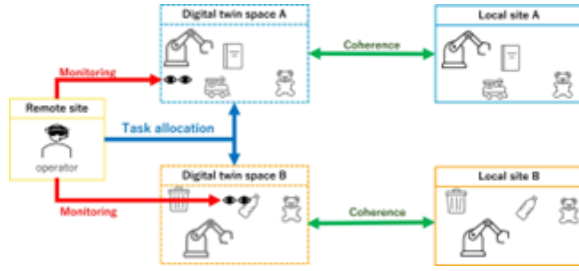


Figure 1: A case of instructing two telepresence robots via the digital twin spaces of the two local sites

3 Experiment on Telecobot

The experimental environment is shown in Figure 2. The local site was a part of our laboratory. Five Lego blocks were set up at the relay point. The experiment was conducted in a scenario in which a robot follows a remote operator's instructions and manipulates items in a nursing home as a local site. The series of tasks in this experiment using Telecobot was as follows.

- Move from a start point to a relay point
- Pinch a piece of Lego block with the robot arm
- Move from the relay point to a target point
- Put the block on the target point

To evaluate the productivity per hour, a time analysis was conducted on the subjects based on video records at the local site during the experiment. Table 1 summarizes the results of the temporal analysis. For the times of operation, preparation, and waiting, the average times required to perform each of the works were measured.

Telecobot required less time for the tasks classified in "operation," resulting in a decrease in time for the remote operator. However, the system may increase the time for the robot to plan and execute the tasks of moving a block in the local site. Telecobot required extra time for the robot to understand the tasks given by the operator based on the point cloud data and to calculate the moving path of the body and arm. The time for the operator to indicate the tasks of the robot by editing the digital twin space was shorter than the waiting time. The result shows the potential of Telecobot in operating more than two robots using the waiting time for the operator.

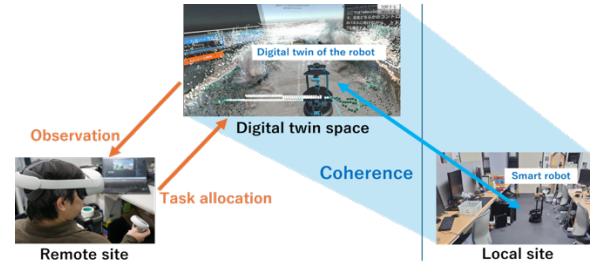


Figure 2: Schematic diagram of the experience, where the remote operator observes the local site via the digital twin space and allocates tasks for the robot onsite by editing the point cloud of the digital twin space. The robot onsite recognizes the difference in the digital twin space and executes the tasks by calculating the moving path of the body and kinematics of the arm.

4 Conclusion

The paper proposed a human-robot telecollaboration system named Telecobot, which has an immersive interface for coordinating multiple telepresence robots remotely. The experiment on the prototype of Telecobot demonstrated that the proposed interface supported a remote operator observing a local site through a digital twin space to build a series of tasks for nursing care. The temporal analysis showed the potential of the system in improving the productivity of multiple robot operation in remote sites. Future prospects include extension of the proposed interface of Telecobot for observing multiple sites to coordinate multiple robots at the same time period via the multiple digital twin spaces.

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Table 1: Temporal analysis of the experiment on Telecobot

	task	Average time
Operation	Move and rotate	22.4
	Pinch and put	21.0
Preparation	Start operation	8.0
	Exit operation	18.5
Waiting	Waiting	38.9
Total		643.6